

PLOT

Plant Layout Organizational Tool

Ergonomic risk mapping
for the plant floor

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Why plant-level risk mapping?

Daily exposure

Repeated loading accumulates across the tasks a worker performs.

Facility decisions

Practitioners need to locate and review assessment results across many work areas.

PLOT links individual assessments to workplace records, spatial maps and group summaries.

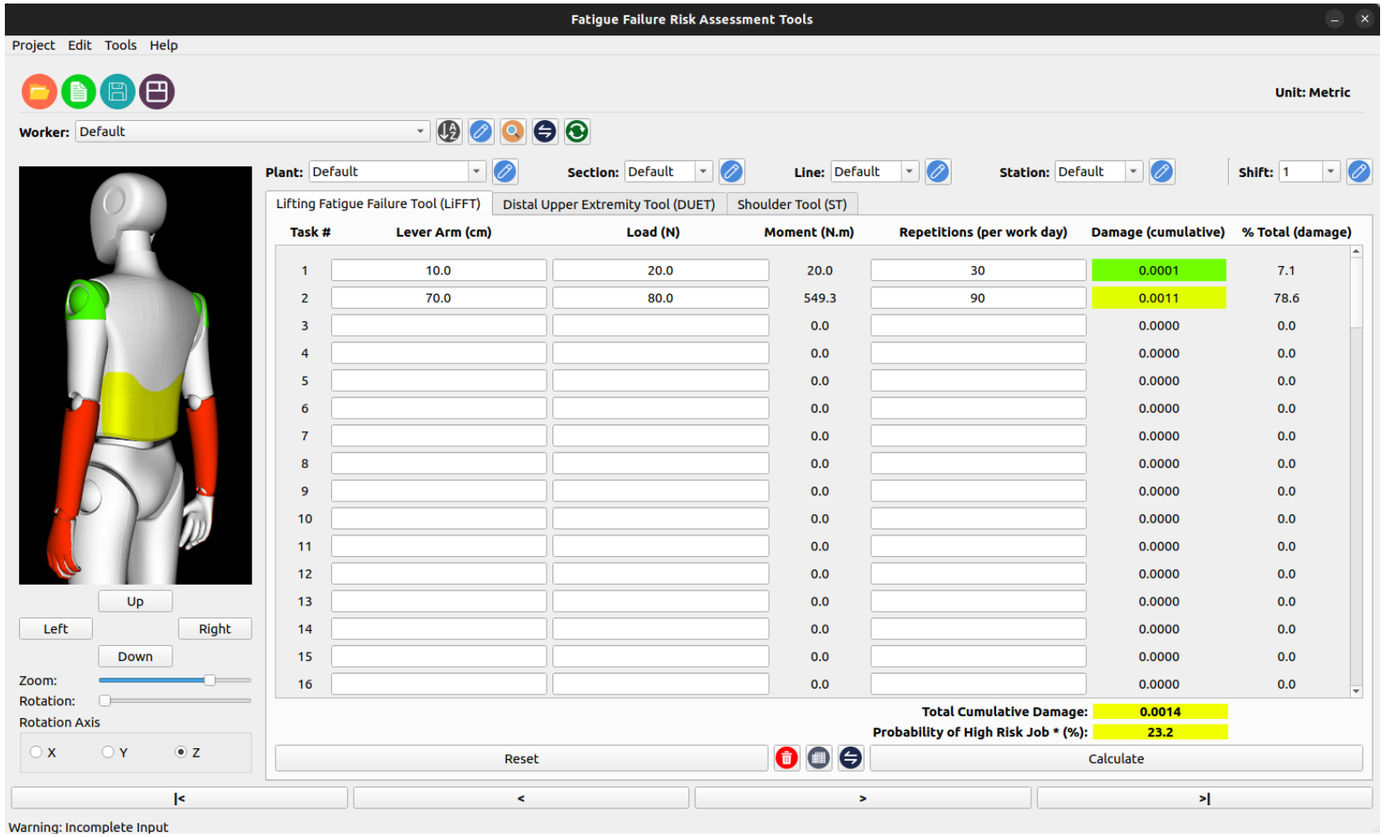
Tools in ErgoTools

One assessment interface for three fatigue failure tools

Tool	Body region	Key task inputs
LiFFT	Low back	Load, horizontal lever arm and repetitions
DUET	Distal upper extremity	Exertion level (OMNI-RES) and repetitions
Shoulder Tool	Shoulder	Task type, load or force, lever arm and repetitions

Outputs: task damage, cumulative damage and tool-specific risk estimates

Initial beta interface



Worker and workplace

A shared context
for each assessment

Three tool tabs

Task inputs and
cumulative results

Project controls

Save records and
open the layout view

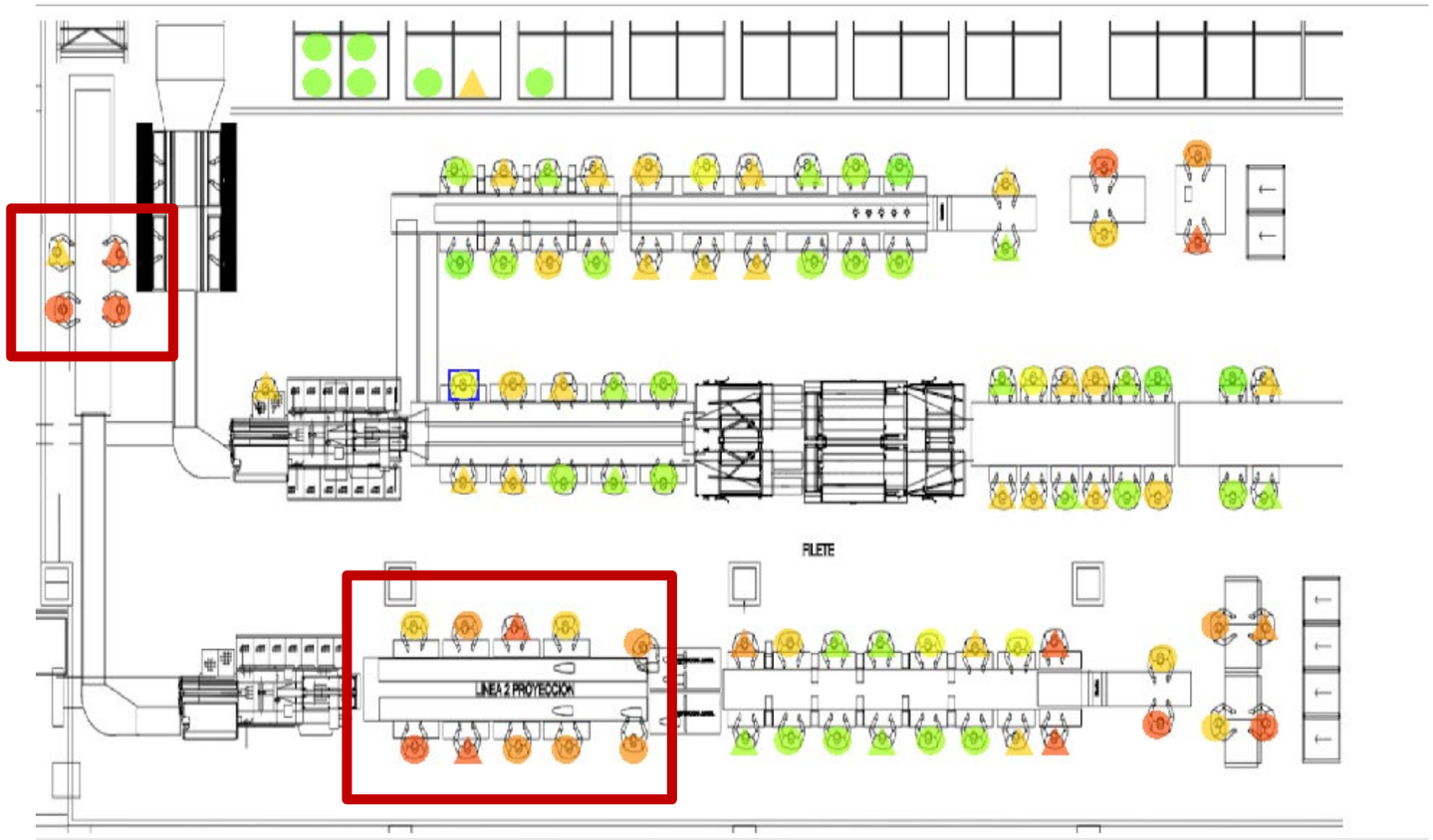
The project database

SQLite links assessments to their workplace context.

Stored structure	Records and relationships
Workplace hierarchy	Plant, section, line and station
Assessment context	Worker linked to a station, shift and tool
Task results	LiFFT, DUET and Shoulder Tool result records

The same records support individual retrieval, workplace filters and group summaries.

Mapped worker assessments



Higher estimated risk: prioritize closer review

Markers connect individual assessment results with locations on the plant floor.

Initial beta layout
96 illustrative worker records

Filters and group summaries



Filter by workplace

Plant, section, line, station and shift

Review selected workers

Individual records and group summaries remain connected.

Review group summaries alongside individual results.

Expert evaluation

4

expert ergonomists

Washington State Department of Labor and Industries

30-minute training session, manuals and short materials

Case	Activities explored
1	Create a worker assessment
2	Manage project records
3	Place workers on a layout
4	Explore a ~100-worker project

SUS scores and open-ended comments informed the redesign.

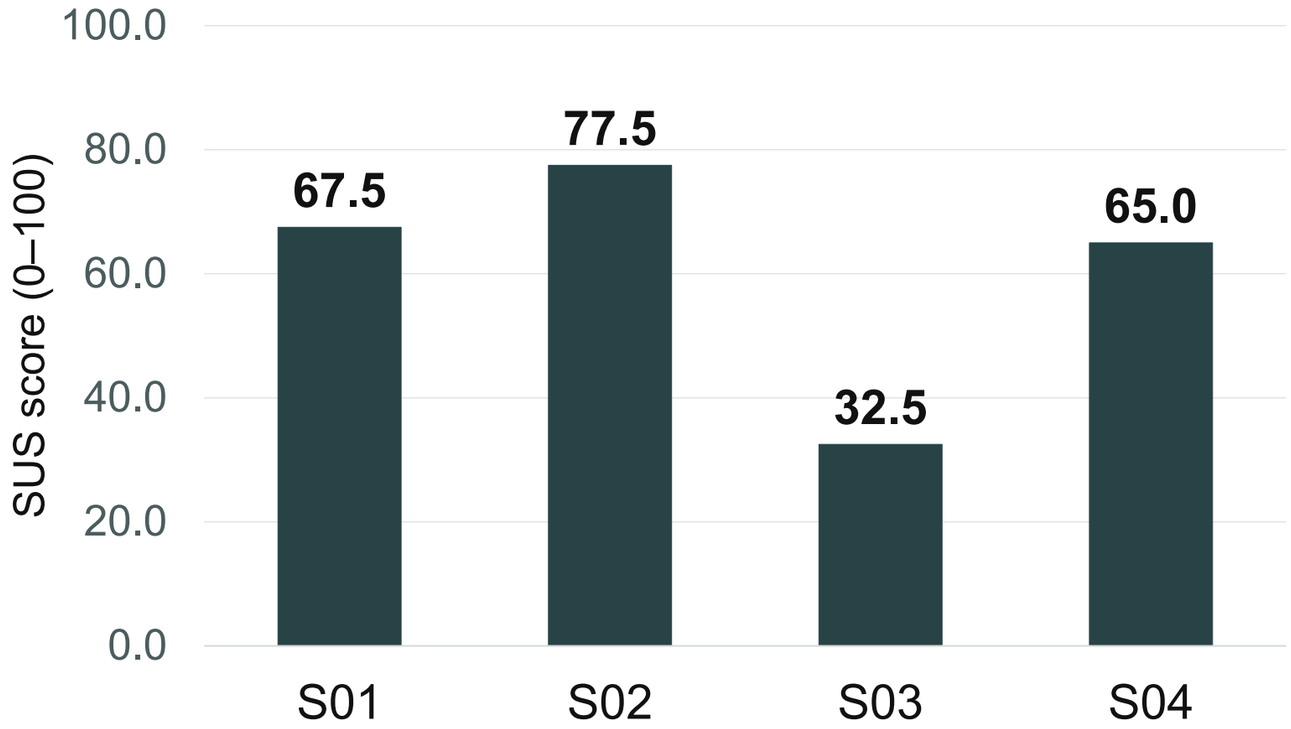
Usability varied across users

Mean SUS score

60.6

± 19.5 SD

Range: 32.5–77.5
Beta version, n = 4



Higher scores indicate better perceived usability. These scores precede the redesign.

Feedback guided the redesign

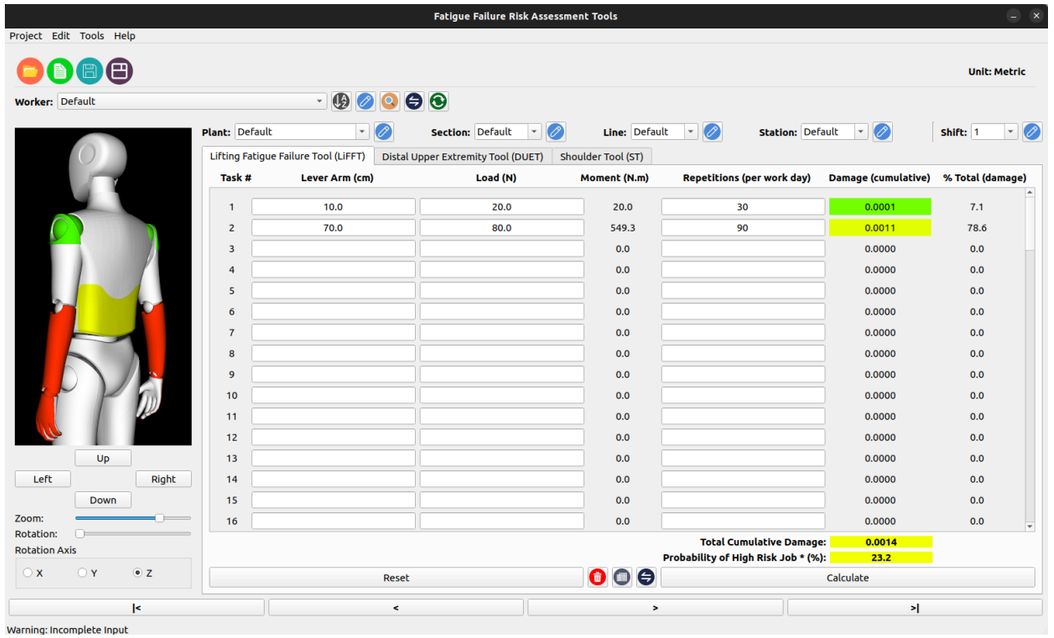
What experts reported	What we changed
Too much information at once	Optional controls collapsed by default
Unclear actions and system state	Labeled icons, tooltips and confirmations
Too many workplace-selection steps	Consolidated workplace selection

“This tool would have been wonderful to use for whole site, multi-job analysis and display of results.”

WSDLI expert user

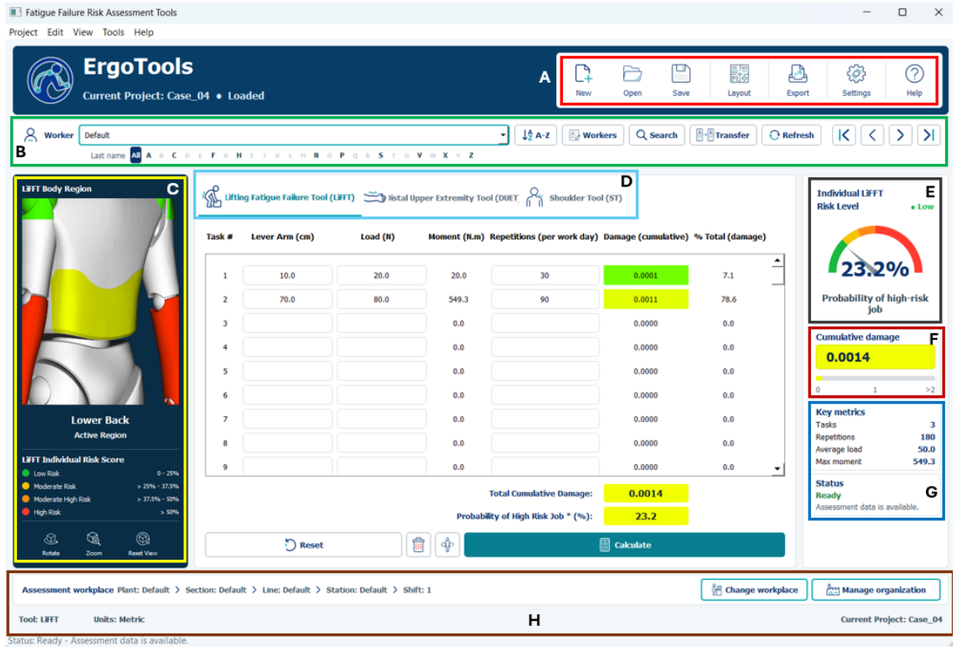
Main interface comparison

Initial beta



Compact icons and distributed controls

Latest version



Labeled actions, results and status

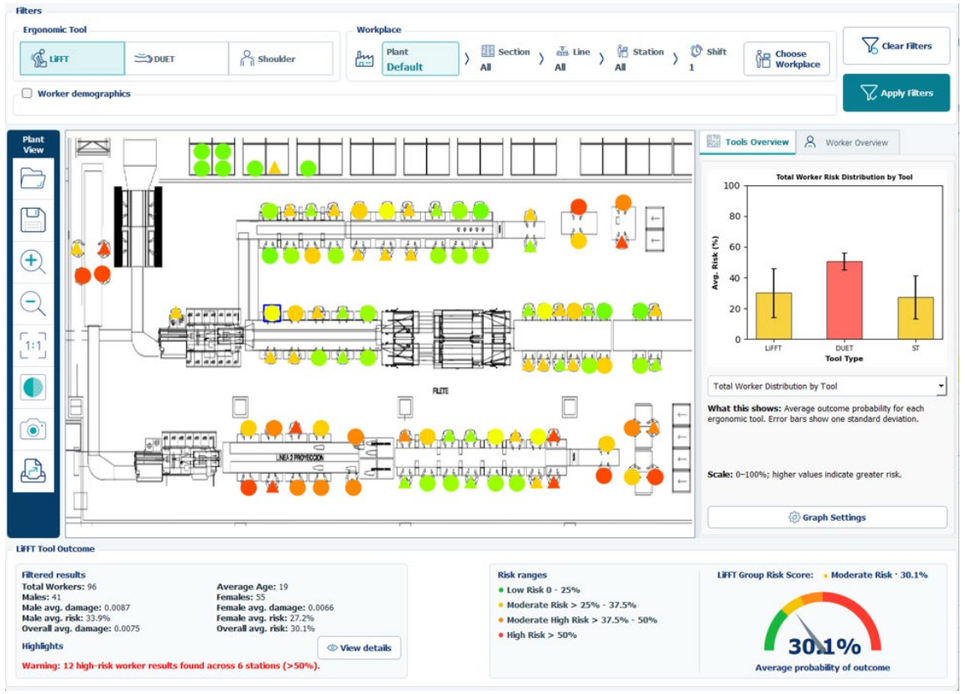
Layout interface comparison

Initial beta



Expanded filters and worker controls

Latest version



Consolidated filters and group results

Lessons learned

Expert users need clear workflows

Training and manuals did not eliminate usability barriers.

Comments made the SUS findings actionable

Specific feedback helped define changes while preserving the core tools.

Next step: evaluate the revised version

Broader user testing and field implementation are still needed.

This four-expert beta evaluation did not establish workplace injury reduction.

PLOT is open source

Individual assessment and facility-level risk mapping in one platform

Source code

http://aub.ie/PLOT_Beta

Thank you to the NSC MSD Solutions Lab and the WSDLI expert evaluators

Project done in collaboration with Universidad Austral de Chile

Contact: Ivan.nail@auburn.edu

Scan for source code

Ergonomics in Design
OnlineFirst
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https://doi.org/10.1177/10648046251408207

Sage Journals

Feature

ErgoTools: Software Platform Integrating Fatigue Failure-Based Risk Assessment Tools for Ergonomics Evaluation in Salmon Processing

Mauricio Henriquez-Schott, Michael Zabala, Sean Gallagher, Susan Kotowski, Michael Jorgensen, Kermit Davis, and Ivan Nail-Ulloa

Proceedings of the Human Factors and Ergonomics Society Annual Meeting
Volume 69, Issue 1, September 2025, Pages 625-632
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https://doi.org/10.1177/10711813251357881

Sage Journals

OE8: Effort and Fatigue

PLOT: A Plant-Layout Organizational Tool Software Based on Fatigue Failure Theory for MSD Control and Mitigation

Mauricio Henriquez-Schott[1], Michael Zabala[2], Robert Sesek[2], Sean Gallagher[2], and Iván Nail-Ulloa[2]



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Thank you so much for your attention!

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